Homework Assignment 11: Pre-Processing

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1 General Instructions

The goal of this assignment is to familiarize you with pre-processing techniques for fairness. You will implement the techniques covered in class on the UCI Adult Census dataset. You will run five different Jupyter notebooks that have been written for you. For each you will answer questions that demonstrate your understanding of the concepts involved. Make sure you review the code carefully and understand how it works. Even though you are not writing code in this assignment, you will be expected to apply some of the techniques to a new dataset during the final assignment.

Each question in the assignment is worth 8 points. The total assignment is worth 200 points.

Begin by downloading the five Jupyter notebooks and UCI dataset into the same directory on your local machine. The files are:

- 1_imputation_examples.ipynb
- 2_oversampling_examples.ipynb
- 3_reweighting_examples.ipynb
- 4_relabeling_examples.ipynb
- 5_proxy_removal_example.ipynb
- adult.data

There are libraries in the notebooks that you will need to install into your Conda environment.

2 Imputation

Run the first notebook, **1_imputation_examples.ipynb**.

1. List the two variables with the highest missingness and provide their percent level of missingness.
2. Are these variables MCAR, MAR, or MNAR with respect to any protected attributes such as sex or race? How do you know?
3. What is the data type for the missing variables? What type of regression will be used to impute them in MICE?

3 Oversampling

Run the second notebook, **2_oversampling_examples.ipynb**. Note that it loads *adult_imputed.csv*, the file created by *1_imputation_examples.ipynb*. The variable *target* represents the income level in dollars of the observation.

1. List percent of total data for each sex x income.
2. What sex x target combinations are underrepresented in the data?
3. What are the new percents of data for each sex x income?

4 Reweighting

Run the third notebook, **3_reweighting_examples.ipynb**. Note that it loads *adult_imputed.csv*, the file created by *1_imputation_examples.ipynb*.

1. The first section of the notebook seeks to reweight to address fairness in a classification model. The model predicts income as a function of the other features. We are concerned about bias with respect to a single protected attribute. What is the name of the variable that is the label in the classification? What is the name of the variable that is the protected attribute?
2. What two reweighting methods are used for the classification?
3. Which model had better performance in terms of accuracy and f1-scores, the unweighted model or the Inverse Prevalence Model? Why do you think this is the case?
4. Which reweighting method narrowed the false positive gap between females and males? Which one narrowed the false negative gap?
5. With classification, we compared FPR and FNR between females and males. With regression, what metric are we comparing?
6. In the second section of the notebook we are trying to predict number of years of education – a regression problem. Did Residual-Parity Weighted Regression narrow the error gap between females and males? Why do you think it did or did not?
7. What would happen if we ran it again? When we would stop repeating it?

5 Relabeling

Run the fourth notebook, **4_relabeling_examples.ipynb**. Note that it loads *adult_imputed.csv*, the file created by *1_imputation_examples.ipynb*. In this notebook, we are running a classification model. We are trying to predict *target* (income level) as a function of the other features. Assume that you work in a bank where the result of the prediction determines whether you offer a loan. You only offer loans to applicants classified as "$> 50k$".

1. Which model has higher performance in terms of accuracy and f1-score? What is the difference in accuracy between them?
2. What proportion of females were scored $> 50k$ in the original model? What about males?

3. What proportion of females were scored $> 50k$ in the relabeled model? What about males?

4. Did error-rate parity get better or worse as a result of relabeling? Why?

5. What type of fairness metric is satisfied by the relabeling?

6 Proxy Removal

Run the fifth notebook, **5_proxy_removal_example.ipynb**. Note that it loads *adult_imputed.csv*, the file created by *1_imputation_examples.ipynb*. This notebook runs two proxy removal routines. The first removes a single attribute, the second removes the combination of two attributes.

1. What is the single attribute removed in the first section?

2. Which feature had the highest absolute value of correlation with the attribute prior to removal? What was the value?

3. Which feature had the highest absolute value of correlation with the attribute after removal? What was the value?

4. What are the attribute removed in the second section?

5. What was the average absolute correlation value to the attributes prior to removal?

6. What was the average absolute correlation value to the attributes after removal?

7. Was there a significant change in the performance of the classification model?